

SOUJANYA S P

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SUMMARY

Data Science enthusiast specializing in generative AI and multi-agent systems, with hands-on experience in building end-to-end machine learning models, agentic AI systems, and data pipelines. Proficient in Python, statistical modeling, and the full ML lifecycle — from data preprocessing and feature engineering to model evaluation and deployment. Skilled in extracting actionable insights from complex datasets and building intelligent solutions that deliver measurable business impact.

PROJECTS

AI Voice Assistant — VAPI · FastAPI · Pandas · SQLite · Streamlit · Agentic Design

- ▶ Built an end-to-end agentic voice assistant that converts unstructured speech inputs into structured database records, automating hospital appointment management and reducing manual data entry by ~70%; records using VAPI and FastAPI, enabling automated hospital appointment management.
- ▶ Engineered a data pipeline and automated reporting module using Pandas and SQLite to capture patient records, reducing reporting time by ~3 hours/week; conversation logs, and scheduling patterns - replacing manual reporting with auto-generated Excel insights.
- ▶ Designed an interactive Streamlit dashboard to visualize real-time doctor availability, appointment trends, and wait time analytics, demonstrating end-to-end data collection, processing, and visualization skills.

Self-Validating Quant Word-Problem Generator - Python · Multi-Agent AI · LLM Orchestration · Agentic Design

- ▶ Architected a production-style multi-agent AI system with dedicated research, problem-generation, dual-solver, and orchestration agents; reduced manual review time to zero across 100% of generated quiz sets — and orchestration agents that autonomously generate and validate quantitative word problems.
- ▶ Implemented hallucination rejection logic and mathematical correctness verification; system produces validated quiz sets with accuracy reports - zero manual review needed.
- ▶ Demonstrates advanced agentic AI design patterns, prompt engineering, and LLM pipeline orchestration.

Movie Recommendation System — Python · Scikit-learn · KNN · Streamlit

- ▶ Developed a personalized movie recommendation engine using K-Nearest Neighbors on large movie datasets delivers efficient, accurate, user-specific suggestions.
- ▶ Designed a modular, production-style codebase with an interactive Streamlit UI; processed and cleaned large datasets for reliable model input.
- ▶ Demonstrates core ML skills: feature engineering, similarity-based learning, model evaluation, and deployment-ready UI integration.

EXPERIENCE

Backend Engineer Intern | Sanskriti Labs

Oct 2025 – Dec 2025 · Bangalore

- ▶ Designed and tested RESTful APIs using NestJS with JWT authentication, middleware handling, and TypeORM for PostgreSQL integration.
- ▶ Applied ETL-style data integration pipelines, improving data flow efficiency by ~40% across structured and unstructured data across modules.
- ▶ Improved backend performance through systematic API debugging, code reviews, and Postman-based testing.

Data Science Intern | Elevate Labs

Apr 2025 – May 2025 · Bangalore

- ▶ Performed exploratory data analysis (EDA) and data preprocessing on real-world business datasets using Python (Pandas, NumPy), extracting actionable patterns and statistical insights to support data-driven decision-making.
- ▶ Created interactive Tableau dashboards to visualize KPIs, enabling delivered recommendations that informed management decisions.
- ▶ Maintained a structured Jupyter Notebook repository with 66+ commits documenting full analysis workflows.

Machine Learning Intern | Cranes Varsity

Mar 2024 – Apr 2024 · Bangalore

- ▶ Completed an intensive 80-hour ML programme implementing supervised and unsupervised learning algorithms in Python.
- ▶ Applied ML techniques to real-world problem-solving projects, gaining hands-on intuition for model training, evaluation, and optimization.

SKILLS

- **Programming Languages:** Python, SQL
- **Data Science & ML:** Machine Learning, Scikit-learn, Pandas, NumPy, OpenCV, PyTorch ,TensorFlow
- **AI / GenAI:** Generative AI, RAG, Multi-Agent Systems, LangChain
- **Frameworks & Tools:** Flask, FastAPI, Streamlit, REST APIs, Git, GitHub, Docker , Tableau , AWS
- **Soft Skills:** Communication, Problem Solving, Analytical Thinking, Teamwork, Quick Learner

EDUCATION

MS RAMAIAH COLLEGE OF ARTS, SCIENCE, AND COMMERCE – Bangalore	2024 – Present
‣ Master of Computer Applications (MCA) Pursuing	
PES INSTITUTE OF ADVANCED MANAGEMENT STUDIES, Shivamogga	2021 – 2024
‣ Bachelor of Computer Applications (BCA) Kuvempu University	
‣ CGPA: 9.38 / 10	
MAHESH PU COLLEGE, Shivamogga	2020 – 2021
‣ Percentage: 98%	

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning Using Python — Cranes Varsity (80-hour certified programme)
- GitHub Essentials — Udemy (version control, collaboration, and open-source workflows)
- AI for Students: Build your own generative model — NxtWave
- Participated in multiple national and college-level hackathons, building AI/ML and full-stack solutions under time constraints.